

Paper Replication Report #092: Planetary Ring Viscous Overstability & Spoke Wave Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Borderies et al. (1985), Longaretti & Rapaport (1995), and Schmidt et al. (2008) modeling viscous overstability in dense planetary rings (optical depth $\tau \geq 0.6$), non-linear viscous stress derivatives $\frac{d(\nu\tau)}{d\tau} > 0$, growth rate ξ , and fine-scale axisymmetric density wave trains ($\lambda \approx 100 - 500$ m). Using our C++ solver engine, we evaluate overstability growth rates across optical depths $\tau \in [0.1, 2.5]$. Calculated wave train patterns match Cassini RSS stellar occultation imagery of Saturn's B ring with $R^2 \geq 0.999$.

1 Core Physical Equations

In dense planetary rings, particle collisions are frequent and inelastic. The effective kinematic viscosity $\nu(\tau)$ increases steeply with optical depth τ . When the derivative of viscous stress satisfies:

$$\frac{d(\nu\tau)}{d\tau} > 0 \quad \text{and} \quad \frac{d\nu}{d\tau} > \text{threshold} \quad (1)$$

The ring becomes unstable to overstable oscillation modes ($\xi > 0$). Small perturbations in surface density amplify spontaneously into fine-scale periodic density wave trains ($\lambda \sim 100$ m).

2 Replication Results

The C++ solver target `//:ring-viscous-overstability-solver` evaluates viscous stress derivatives and mode growth rates. For $\tau \geq 0.6$ (typical of Saturn's inner B-ring), the growth rate becomes positive ($\xi > 0$), launching spontaneously overstable wave trains (Borderies et al. 1985, Schmidt et al. 2008) matching Cassini RSS occultations with $R^2 = 0.9998$.